B. Even Modulo Pair

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1000 | Tags: brute force, math, number theory | https://codeforces.com/problemset/problem/2164/B

You are given a **strictly increasing** sequence of positive integers a_1 lt a_2 lt ... lt a_n. Find two distinct elements x and y from the sequence such that x lt y and y mod x is even, or determine that no such pair exists.

p mod q denotes the remainder from dividing p by q.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 2 \cdot 10^4$). The description of the test cases follows.

The first line of each test case contains one integer n ($2 \le n \le 10^5$) — the length of the sequence.

The second line of each test case contains n integers $a_1, a_2, \ldots, a_n$ ($1 \le a_1$ lt ... lt $a_n \le 10^9$) — the given sequence.

It is guaranteed that the sum of n over all test cases does not exceed 10^5.

Output

For each test case:

- If no such pair exists, output -1.
- Otherwise, output two integers x and y — the elements that satisfy the condition.

If there are multiple valid pairs, you may output any of them.

Example

Input

```
4
5
1 3 4 5 6
6
2 3 5 7 11 13
4
2 3 13 37
3
17 117 1117
```

Output

```
3 5
3 11
-1
17 1117
```

Note

Visualizer link

In the first test case, choosing $x = 3$ and $y = 5$ yields $y \bmod x = 5 \bmod 3 = 2$, which is even.

In the third test case, it is clear that no valid pair exists.